Lab 3: Architecture Justification

PES1UG24CS058 - Anagha N | Smart Lab Equipment & Slot Reservation Portal

We chose Layered Client-Server Architecture for the Smart Lab Equipment & Slot Reservation Portal.

Architectural Choice. A three-tier layered client-server architecture: a Web Client (presentation), server-side components behind interfaces (Reservation Manager, Equipment Service, Auth & Access Control, Notification Adapter) and a Data Access Layer over one relational database. Microservices were rejected because the portal serves a single lab and splitting reservation, calibration and data into separate deployable services would add network calls and distributed transactions for no benefit. A plain client-server design without layers was rejected because reservation rules would mix with UI and database code and be hard to test.

Reason 1: Consistency under concurrent bookings. NFR-003 requires that when several requests arrive for the same equipment and overlapping slot, at most one is confirmed, using database-level lock isolation. With one server and one relational database reached only through the Data Access Layer, the overlap check and the slot lock run in a single transaction. In a microservices design the same guarantee would need distributed transactions.

Reason 2: One responsibility per component. Each rule in the SRS has one home: slot rules, overlap checks and tokens in the Reservation Manager (FR-001, FR-010), calibration status and returns in the Equipment Service (FR-002, FR-004, FR-008), login and roles in Auth & Access Control (FR-006, NFR-002). Components talk only through the interfaces IReservation, IEquipment, ICalibration, INotify, IData and IAuthenticate, so each can be changed or tested on its own. This also matches the small scale and simple deployment of a single-lab system.

Security advantage. Every request enters the server, where Auth & Access Control enforces role-based access: only a Lab Technician can update calibration status, manage equipment or record returns, and a session token expires after 30 minutes of inactivity (NFR-002). Clients never reach the database directly, so there is one place to enforce and audit access rather than several.

Performance benefit. A reservation takes a short synchronous path (client, server, database) with no hops between services, which supports the target of 200 ms or less per request at 20 concurrent requests (NFR-001). The confirmation notification is sent through the Notification Adapter to the external system, so a slow notification does not delay the reservation response.